SUMMER INTERNSHIP REPORT

AWS CLOUD & SERVERLESS APPLICATION DEVELOPMENT

REPORT SUBMITTED TO

AMITY UNIVERSITY MAHARASHTRA

FOR THE DEGREE OF

MASTER OF COMPUTER APPLICATIONS (MCA)

BY

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AMITY UNIVERSITY MAHARASHTRA

September 2026

AMITY UNIVERSITY MAHARASHTRA

Established vide Maharashtra Act No.13 of 2014, of Government of Maharashtra, and recognized under Section 2 (f) of UGC Act 1956.

CERTIFICATE

This is to certify that the summer internship report titled “AWS Cloud & Serverless Application Development” submitted by Narayan Behera, Enrollment No. A710145025032, in partial fulfilment of the requirements of the degree of Master of Computer Applications (MCA) at Amity University Maharashtra, is based on the work carried out during his internship at F13 Technologies as an AWS Cloud Intern during May 2026 to August 2026.

The report presents the technical work, cloud services, projects, troubleshooting activities, learning outcomes, and professional experience gained during the internship.

Place: Mumbai Signature of Student

Date: _____

Faculty Guide: _____

Signature of Guide

Prof. (Dr.) Manoj Devare
Professor & HOI-AIIT
Director, Accreditation and Ranking-AUM

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I would like to express my sincere gratitude to F13 Technologies for providing me with the valuable opportunity to undertake an internship as an AWS Cloud Intern. This internship has been an enriching and rewarding experience that provided me with practical exposure to cloud computing, AWS services, serverless architecture, cloud-based application development, service integration, application deployment, monitoring, troubleshooting, and professional software development practices. The knowledge and hands-on experience gained during this period have significantly enhanced my technical understanding and helped me connect my academic learning with real-world applications.

I am deeply thankful to my internship mentor, Kabir, for his continuous guidance, technical support, valuable feedback, and encouragement throughout the internship. His mentorship and practical insights helped me understand various AWS concepts and approaches to solving technical challenges. His guidance played an important role in improving my problem-solving abilities and developing a better understanding of cloud-based technologies and professional work practices.

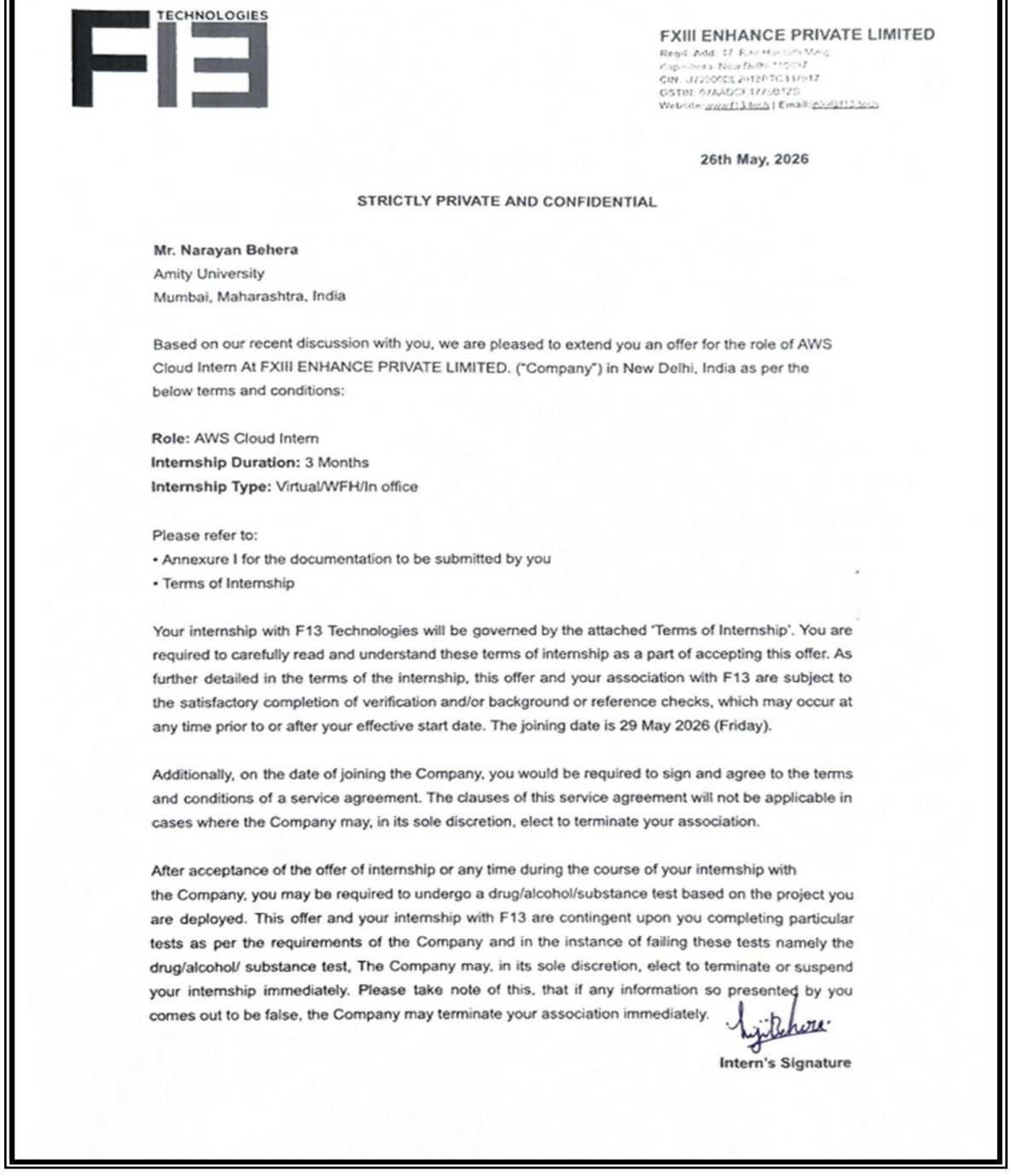
I would also like to express my heartfelt gratitude to my internship guide, Dr. C. Kalpana, for her continuous support, guidance, encouragement, and valuable suggestions throughout the course of my internship. Her mentorship was instrumental in helping me stay focused on my learning objectives and successfully complete the internship and its documentation. Her academic guidance and encouragement have contributed significantly to my overall learning and professional growth.

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industry learning.

Finally, I would like to thank my classmates, colleagues, friends, and everyone who directly or indirectly supported me during the internship and while preparing this report. Their encouragement, assistance, and positive support helped me overcome challenges and complete this internship successfully. I am sincerely grateful to everyone who contributed to making this learning experience meaningful and valuable.

Internship Offer Letter



TECHNOLOGIES

FXIII ENHANCE PRIVATE LIMITED
Regd. Add: 17, East Hanuman Marg,
Kapashera, New Delhi-110037
CIN: U72900DL2018PTC337917
GSTIN: 07AADCF7758J1ZS
Website: www.f13.tech | Email: pvt@f13.tech

26th May, 2026

STRICTLY PRIVATE AND CONFIDENTIAL

Mr. Narayan Behera
Amity University
Mumbai, Maharashtra, India

Based on our recent discussion with you, we are pleased to extend you an offer for the role of AWS Cloud Intern At FXIII ENHANCE PRIVATE LIMITED. ("Company") in New Delhi, India as per the below terms and conditions:

Role: AWS Cloud Intern
Internship Duration: 3 Months
Internship Type: Virtual/WFH/In office

Please refer to:
• Annexure I for the documentation to be submitted by you
• Terms of Internship

Your internship with F13 Technologies will be governed by the attached 'Terms of Internship'. You are required to carefully read and understand these terms of internship as a part of accepting this offer. As further detailed in the terms of the internship, this offer and your association with F13 are subject to the satisfactory completion of verification and/or background or reference checks, which may occur at any time prior to or after your effective start date. The joining date is 29 May 2026 (Friday).

Additionally, on the date of joining the Company, you would be required to sign and agree to the terms and conditions of a service agreement. The clauses of this service agreement will not be applicable in cases where the Company may, in its sole discretion, elect to terminate your association.

After acceptance of the offer of internship or any time during the course of your internship with the Company, you may be required to undergo a drug/alcohol/substance test based on the project you are deployed. This offer and your internship with F13 are contingent upon you completing particular tests as per the requirements of the Company and in the instance of failing these tests namely the drug/alcohol/ substance test, The Company may, in its sole discretion, elect to terminate or suspend your internship immediately. Please take note of this, that if any information so presented by you comes out to be false, the Company may terminate your association immediately.

Intern's Signature

TECHNOLOGIES

FXIII ENHANCE PRIVATE LIMITED

Regd. Add: 37, Kasturba Marg,
Rajendra Nagar, New Delhi 110037
CIN: U72900DL2019PTC357917
GSTIN: 07AADCF1775B1ZS
Website: www.f13.tech | Email: studio@13.tech

The company may also ask you to attend a certain set of AWS modules and attain certificates for the same, the cost of the same shall be borne by you and won't be refundable or transferable.

In the event a government body/authority exercising its jurisdiction and statutory power/authority seeks information pertaining to any aspect of your association, the Company shall provide such information to the government body/authority without any notification to you. The foregoing shall be applicable to information pertaining to your internship being shared in pursuance of statutory requirements/compliance. You may belong to this category and your details will be disclosed to these authorities.

To indicate your acceptance of this offer and Terms of Internship with F13, please confirm your acceptance/rejection by signing a copy of this and revert back to us within the timeframe as directed by the HR team. If we do not receive your response within the timeframe suggested by the HR team, the terms of this Internship Offer will be deemed to have been rejected by you, unless otherwise communicated to you by the Company in writing.

Further, at the time of joining you are required to provide all documentation identified in Annexure-II along with a signed copy of this offer letter, Non-Disclosure Agreement and Terms of Internship.

We look forward to hearing from you. We wish you a successful career ahead of you and look forward to you joining us.

Authorised Signatory
FXIII ENHANCE PVT LTD

Intern's Signature

FXIII ENHANCE PVT. LTD. F13 TECHNOLOGIES

Internship Completion Letter



CERTIFICATE
OF COMPLETION

PROUDLY PRESENTED TO

Narayan Behera

On Successful Completion of his tenure as an

AWS Cloud Intern

He enthusiastically participated in all the assigned tasks and demonstrated his admirable work ethic, research and leadership skills.

We at F13 Technologies have been delighted to have him as our Intern. We thank him for helping us make changes happen. We wish him all the best for his future endeavours.

FXIII ENHANCE PVT. LTD. NEW DELHI F13 TECHNOLOGIES

Kartik Kharbanda
Co-founder & Director

F13 TECHNOLOGIES
FXIII ENHANCE PRIVATE LIMITED

SCAN TO VALIDATE

Cert no: f1b139e

29th Aug 2026

New Delhi - IN

CHAPTER 1 – INTRODUCTION

1.1 Background of Internship

Cloud computing has become a fundamental part of modern software development and IT operations. My internship at F13 Technologies provided practical exposure to Amazon Web Services (AWS), serverless application development, cloud storage, computing, APIs, databases, event-driven services, monitoring, permissions, and deployment workflows.

The internship experience helped me connect academic concepts with practical implementation. During the internship, I worked with various AWS services, including Amazon EC2, Amazon S3, AWS Lambda, Amazon API Gateway, Amazon DynamoDB, Amazon SNS, Amazon EventBridge, Amazon Rekognition, Amazon SES, Amazon CloudWatch, and AWS IAM. Working with these services provided practical understanding of how different cloud components can be integrated to develop, deploy, monitor, and manage cloud-based applications.

1.2 Objectives of Internship

The primary objective of the internship was to understand AWS cloud infrastructure and appropriate service selection based on application requirements. The internship focused on gaining practical experience with different categories of cloud services, including computing, storage, serverless computing, APIs, databases, event-driven services, identity and access management, and monitoring.

Another important objective was to build serverless workflows by integrating services such as S3, API Gateway, Lambda, and DynamoDB. The internship also provided an opportunity to explore a facial-recognition-based attendance system using S3, Rekognition, Lambda, DynamoDB, API Gateway, and EventBridge.

The internship further aimed to develop practical troubleshooting skills by working with common cloud-related issues, including IAM AccessDenied errors, S3

permission problems, Lambda runtime and dependency issues, API integration problems, and HTTP 403 errors. In addition, the internship focused on understanding cloud security, least-privilege access, monitoring, and reliable deployment practices.

1.3 Duration & Location

The internship was carried out from May 2026 to August 2026 in a remote working mode with F13 Technologies. The remote format provided an opportunity to work with cloud-based tools and services while following a practical development and troubleshooting workflow.

1.4 Methodology Followed

The methodology followed during the internship involved first studying AWS service concepts and understanding their practical use cases. Cloud resources were then created incrementally, with each component validated independently before being integrated with other services. Different AWS services were connected through APIs, triggers, permissions, and event flows to create functional cloud workflows.

The developed applications and workflows were tested systematically by inspecting logs and service responses, isolating errors, and applying appropriate corrective changes. Troubleshooting was an important part of the methodology, particularly while dealing with permissions, API integration, Lambda execution, and service configuration issues. The architectures, resources, troubleshooting steps, and learning outcomes were also documented throughout the internship to maintain a clear record of the implementation process.

1.5 Important Points

The internship emphasized practical implementation rather than only theoretical study. The work involved building cloud workflows, integrating multiple AWS services, debugging access and integration issues, and understanding the practical use of serverless architectures. Through this experience, I gained a better understanding of how serverless architectures can reduce infrastructure management requirements for application developers while enabling the integration of different cloud services to develop functional and scalable application workflows.

CHAPTER 2 – COMPANY PROFILE

2.1 Company History & Background

F13 Technologies provided the internship environment in which I worked on AWS and cloud-oriented development activities. The internship exposure was centered on practical software and cloud engineering tasks, with emphasis on understanding how managed AWS services can be combined to build functional application workflows.

During the internship, the practical learning environment focused on working with different AWS services and understanding their roles within cloud-based application development. This provided an opportunity to gain hands-on experience with cloud technologies while applying software development concepts in practical scenarios.

2.2 Mission, Vision, and Values

The internship experience highlighted several values that are important in professional technical work, including continuous learning, practical problem-solving, reliable implementation, security-conscious configuration, teamwork, and clear documentation. These values were reflected throughout the internship through the iterative process of building, testing, troubleshooting, and improving cloud-based solutions.

The experience also emphasized the importance of approaching technical challenges systematically, understanding the underlying cause of an issue, implementing appropriate solutions, and documenting the results clearly. This helped develop a professional approach toward cloud engineering and software development activities.

2.3 Products / Services – Internship Exposure

During the internship, the major areas of practical exposure included cloud and AWS-oriented application development, with a focus on using managed cloud services for application workflows. The internship provided experience with serverless application architecture and API integration, including the integration of different AWS services to support application functionality.

The internship also involved cloud storage and database integration, along with the development of event-driven application workflows using appropriate cloud services. In addition, practical exposure was gained in cloud monitoring and operational troubleshooting, which involved identifying and resolving configuration, permission, and integration-related issues.

Another important area of exposure was application deployment and configuration using managed cloud services. This helped in understanding how cloud resources can be configured and integrated to support application development and deployment requirements.

2.4 Industry Positioning

The internship was aligned with the cloud-computing and software-development industry, where organizations increasingly use managed cloud services, serverless platforms, APIs, event-driven architectures, and cloud-native monitoring to reduce infrastructure overhead and accelerate application delivery.

The practical exposure provided during the internship helped me understand the role of cloud technologies in modern software development. Working with AWS managed services demonstrated how different cloud components can be integrated to create application workflows while reducing the need for extensive infrastructure management. This experience also provided an understanding of the importance of scalability, security, monitoring, and reliable configuration in cloud-based application development.

CHAPTER 3 – INTERNSHIP DETAILS

3.1 Nature of Work Assigned

My internship role at F13 Technologies was AWS Cloud Intern. The work involved learning and applying AWS services to practical application scenarios. During the internship, I worked with cloud computing, object storage, serverless functions, API management, NoSQL databases, event-driven services, identity and access management, monitoring, and application troubleshooting.

A major part of the internship involved understanding complete request and service flows. For example, a web application could send data to an AWS endpoint, where API Gateway would receive and route the request to Lambda for processing, and DynamoDB could then be used to store the resulting data. Understanding this complete chain helped me identify configuration, integration, and permission-related issues more effectively.

3.2 Daily / Weekly Tasks

Week 1 – AWS Fundamentals and Environment Setup

During the initial phase, I reviewed core AWS services and their practical use cases. I worked with IAM roles and permissions and created and tested cloud resources using a structured troubleshooting approach. I also studied the differences between traditional compute and serverless execution to understand when different approaches could be used in cloud application development.

Week 2 – Serverless Contact Form

In the second phase, I developed a web-based Serverless Contact Form hosted using Amazon S3. I configured API Gateway as the public API endpoint and implemented an AWS Lambda function to process the submitted form data. The submitted messages were stored in DynamoDB for persistent storage.

The complete S3 → API Gateway → Lambda → DynamoDB workflow was tested to ensure that data could successfully move through each component. During this phase, I also troubleshooted CORS, IAM, API integration, and permission-related issues encountered during development and testing.

Week 3 – Facial Recognition Attendance

During the third phase, I worked on a Facial Recognition Attendance System. Amazon S3 was used for image storage, while Amazon Rekognition was used for facial analysis. Lambda functions were used for registering and recognizing employee faces, while DynamoDB was used to store employee and attendance information.

The system also involved integrating API Gateway and EventBridge to support application communication and event-driven workflows. During implementation and testing, I investigated access permissions, event triggers, and service integration issues to ensure that the individual components worked correctly together.

Week 4 – Supporting Cloud / LMS Work

In the supporting cloud and LMS-related phase, I worked with services including Amazon SES, Amazon SNS, CloudWatch, and EventBridge. I explored certificate-generation and certificate-storage workflows and gained additional experience with serverless application components.

I also practiced application monitoring and debugging using CloudWatch logs. This helped me understand how execution logs can be used to identify errors, trace application behavior, and troubleshoot issues in distributed cloud workflows.

3.3 Tools, Software, and Methods Used

The internship involved working with a range of AWS services and development tools. Amazon EC2 was used for understanding cloud computing and virtual-server concepts. Amazon S3 was used for static website hosting, object storage, and image or document storage. AWS Lambda was used for event-driven and serverless application logic. Amazon API Gateway was used for HTTP and API integration between web clients and Lambda functions, while Amazon DynamoDB provided NoSQL persistence for application data. Amazon Rekognition was used for facial analysis in the attendance use case, and Amazon EventBridge was used for event-driven integration and scheduled or event-based workflows.

Other services included Amazon SNS and Amazon SES for notification and email-related workflows, AWS CloudWatch for logging and monitoring, and AWS IAM for managing identities, roles, policies, and least-privilege access.

For source-code management and project versioning, I worked with Git and GitHub. For web interfaces connected to cloud backends, I worked with React, JavaScript, HTML, and CSS. These tools and services together provided practical exposure to the development, deployment, integration, monitoring, and troubleshooting of cloud-based applications.

3.4 Process Followed

The development process followed a modular and systematic approach. The process began by defining the application requirement and selecting the appropriate AWS services. The required cloud resources were then created and configured, followed by the setup of appropriate permissions and IAM policies.

After the individual services were configured, they were connected through APIs, triggers, and event flows. The integrations were then tested to verify the expected behavior. When an error occurred, logs and responses were inspected to identify the source of the problem, after which the required configuration or code changes

were applied.

The working architecture can be represented as:

Web Client → API Gateway → Lambda → DynamoDB / S3

CHAPTER 4 – PROJECT / WORK DONE

4.1 Serverless Contact Form

The Serverless Contact Form was designed as a web application in which users can submit their contact information through a browser. The project demonstrates how a frontend application can communicate with a serverless backend without requiring a permanently running application server. The architecture of the application follows the flow S3 → API Gateway → Lambda → DynamoDB, where each AWS service performs a specific role in processing and storing the submitted information.

Step-by-Step Methodology

The implementation began with creating and configuring an S3-hosted frontend for the contact form. A DynamoDB table was then created to store the submitted contact messages. A Lambda function was developed to receive, validate, and process the submitted payload before storing the required information in DynamoDB.

Next, an API Gateway route was created and connected to the Lambda function. Appropriate IAM permissions were configured to allow the Lambda function to write data to DynamoDB. The frontend was then configured to communicate with the API Gateway endpoint. Finally, both successful and unsuccessful requests were tested, and the resulting execution information and errors were inspected through CloudWatch.

Key Learning

The project demonstrated that a small application can be implemented using managed cloud services, with each component having a focused responsibility. It provided practical understanding of API contracts, JSON payloads, IAM permissions, CORS, Lambda execution, and DynamoDB persistence. The project

also helped demonstrate how multiple AWS services can work together to create a complete serverless application workflow.

Major Troubleshooting Areas

During the implementation, several troubleshooting areas were encountered. These included HTTP 403 and authorization failures, which required checking the API and access configuration. IAM AccessDenied errors were investigated by reviewing whether the required permissions had been granted to the appropriate execution role. In addition, S3 permission and hosting configuration issues were addressed by checking bucket settings and access permissions.

4.2 AWS LMS / Supporting Cloud Work

Alongside the main projects, I worked on supporting cloud functionality for an LMS-style application. The work involved AWS Lambda functions, S3-based certificate storage, and automated certificate-generation workflows. This provided additional practical experience in designing serverless components and connecting them with persistent cloud storage.

The work involved creating and testing Lambda functions for LMS operations and using Amazon S3 to store generated certificates. I also worked with Python-based serverless logic and document generation, which provided practical experience in implementing backend functionality using Lambda. During deployment, runtime and library-related issues were encountered and troubleshooted by checking compatibility, dependencies, and deployment structure. Generated outputs were tested and validated, while cloud-resource permissions were also checked to ensure that the required services could interact correctly.

4.3 Challenges Faced and Solutions

Several technical challenges were encountered while developing and testing the cloud-based applications. IAM AccessDenied errors were addressed by reviewing Lambda execution roles and verifying that the required AWS service actions were included in the associated permissions. S3 access errors were investigated by checking bucket configuration, object access, and IAM permissions.

During API Gateway integration, routes, integrations, deployment stages, request and response behavior, and CORS configuration were verified to identify integration problems. Lambda runtime and dependency issues were addressed by checking runtime compatibility, required packages, and the deployment structure.

For distributed workflow debugging, CloudWatch logs were used to trace execution and identify errors, while individual components were tested independently to isolate problems. When working with event-driven workflows, the event sources and targets were validated to ensure that EventBridge events were being configured and processed correctly.

Overall, these challenges helped develop a systematic approach to cloud troubleshooting. Instead of checking the entire application at once, individual AWS components, configurations, permissions, inputs, outputs, and logs were examined to identify and resolve issues effectively.

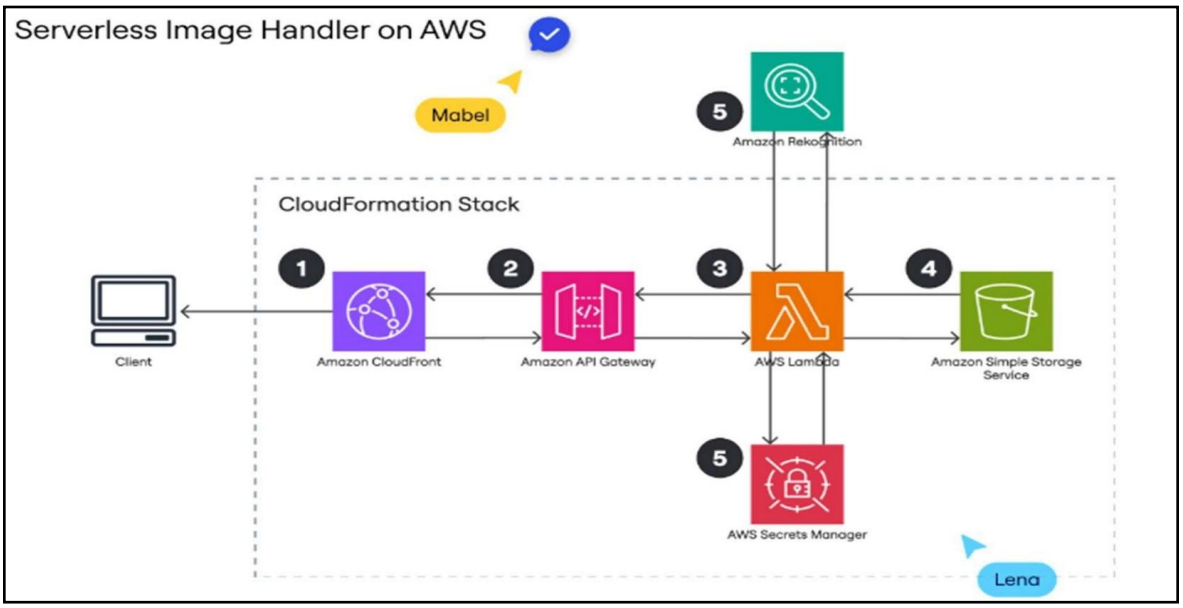


Fig 1.1 Serverless Image Handler on AWS

CHAPTER 5 – LEARNING & OUTCOMES

5.1 Technical Skills Gained

During my internship at F13 Technologies, I developed practical knowledge of various AWS services covering areas such as compute, storage, serverless computing, APIs, databases, AI services, event-driven services, messaging, identity and monitoring. The internship provided me with a better understanding of how different AWS services can be integrated to develop and manage cloud-based applications.

I gained a strong understanding of serverless architecture and Lambda-based application development. I also developed hands-on knowledge of Amazon S3, including buckets, objects, hosting, and cloud storage. Working with Amazon API Gateway provided practical experience in HTTP-based API integration, while working with Amazon DynamoDB helped me understand how database-backed workflows can be designed and implemented.

The internship also introduced me to Amazon Rekognition integration and its application in cloud-based solutions. I gained practical knowledge of AWS IAM, including roles, policies, permissions, and the principle of least-privilege access. In addition, I used Amazon CloudWatch for troubleshooting and log analysis, which helped me understand the importance of monitoring in cloud applications. I also gained experience with Git and GitHub for source-code management and version control.

5.2 Soft Skills Developed

The internship helped me develop several important professional and soft skills. I improved my structured problem-solving and debugging abilities by systematically identifying and resolving technical issues. I also developed better technical documentation and communication skills through the documentation of cloud architectures, implementation steps, troubleshooting processes, and learning outcomes.

Managing multiple cloud-related tasks helped me improve my time-management skills and ability to organize work effectively. I learned to break complex distributed systems into smaller components while troubleshooting, making it easier to identify the source of problems. The internship also encouraged independent learning, particularly when working with unfamiliar AWS services and concepts.

Working in a remote environment further helped me develop professional collaboration skills, including clear communication, task coordination, and maintaining consistent progress while working on cloud-based assignments.

5.3 Industry Exposure

The internship provided valuable exposure to practical cloud-engineering expectations. I learned that cloud development is not limited to simply creating and configuring cloud resources. Successful cloud-based applications also require appropriate permissions, secure configuration, effective API design, monitoring, testing, documentation, and systematic troubleshooting.

The practical experience helped me understand how these different aspects work together in a real-world cloud environment. It also provided an understanding of the importance of reliability, security, and proper configuration when developing and deploying applications using managed cloud services.

5.4 Academic Concepts Applied in Work

The internship provided an opportunity to apply several academic concepts to practical cloud-development tasks. Database Management concepts were applied while working with DynamoDB for data modelling and persistence. Computer Networks concepts were applied through HTTP APIs, API Gateway, CORS, and communication between different cloud services.

Concepts related to Operating Systems and system-level computing were applied while understanding compute resources, runtimes, processes, and deployment concepts. Software Engineering principles were applied through modular design, testing, debugging, documentation, and version control.

Programming concepts were also applied during the internship through JavaScript and React-based frontend work and Python-based serverless logic. This combination of academic knowledge .

CHAPTER 6 – CONCLUSION

6.1 Summary of Internship Experience

The internship at F13 Technologies was a valuable and practical learning experience focused on AWS cloud computing and serverless application development. As an AWS Cloud Intern, I gained hands-on exposure to a broad range of AWS services and developed an understanding of how these services can be integrated to implement complete and functional application workflows.

The Serverless Contact Form project strengthened my practical understanding of Amazon S3, API Gateway, AWS Lambda, DynamoDB, IAM, and CORS. Working on this project helped me understand how frontend applications can communicate with backend services through APIs and how serverless components can be integrated to process and store application data.

The Facial Recognition Attendance System further expanded my understanding by introducing Amazon Rekognition and event-driven integration. This experience provided practical knowledge of combining multiple AWS services to implement a cloud-based workflow involving facial recognition, data processing, and attendance management. Supporting LMS-related work further improved my ability to build, deploy, configure, and troubleshoot serverless components.

One of the most valuable outcomes of the internship was learning a systematic approach to debugging cloud applications. Instead of treating a cloud application as a single large system, I learned to trace requests through individual services and verify their configuration, permissions, inputs, outputs, and logs. This approach improved my ability to identify the root cause of technical issues and apply appropriate solutions.

Overall, the internship helped bridge the gap between academic knowledge and practical cloud engineering. It strengthened my technical skills, problem-solving

abilities, and understanding of real-world AWS-based application development.

6.2 Future Plan

Going forward, I plan to further deepen my knowledge and practical skills in cloud engineering and DevOps, particularly in areas such as AWS architecture, Linux, Docker, Kubernetes, CI/CD, infrastructure automation, monitoring, and secure cloud deployment. I also intend to strengthen my understanding of production-grade architecture, observability, cost optimization, and Infrastructure as Code (IaC).

The internship has strengthened my interest in pursuing a career in CloudOps, DevOps, and software engineering. The practical exposure gained at F13 Technologies has provided a strong foundation for developing more advanced cloud-native applications, designing scalable architectures, and implementing automated deployment pipelines. In the future, I aim to continue building my technical expertise through practical projects and industry-oriented learning while applying cloud and DevOps practices to real-world software development challenges.

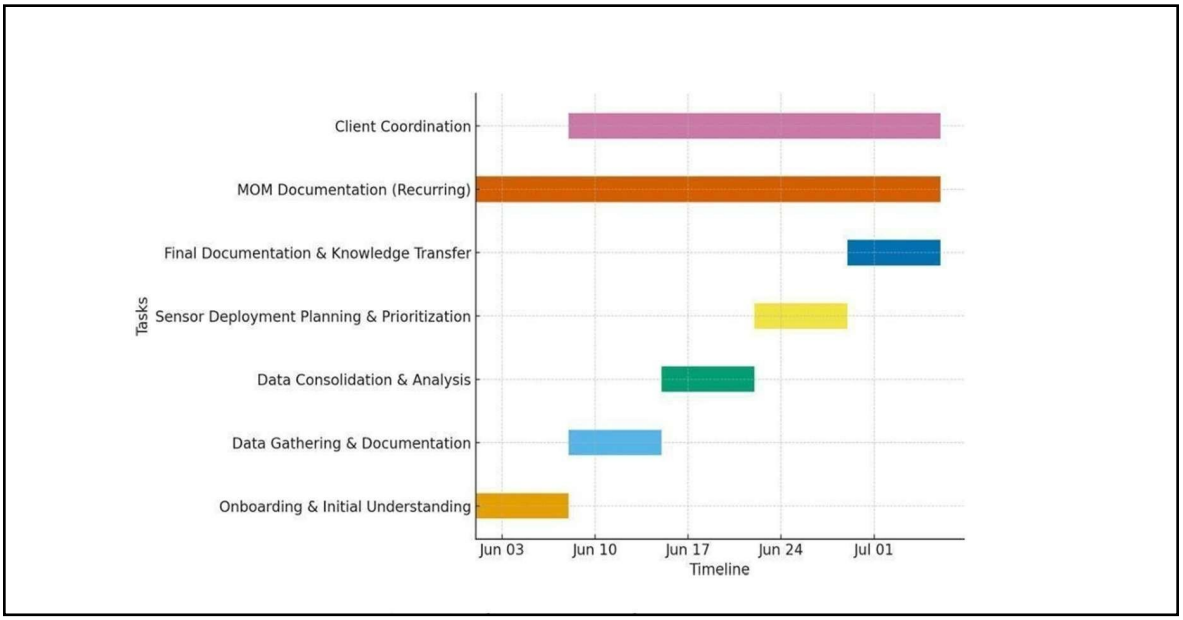


Fig 1.2 Timeline – Gantt Chart Overview

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APPENDICES

Appendix A – Serverless Contact Form Architecture

The Serverless Contact Form follows a serverless architecture in which the frontend is hosted on Amazon S3 and communicates with the backend through Amazon API Gateway. The request is processed by AWS Lambda, and the submitted contact information is stored in Amazon DynamoDB.

Architecture Flow:

Web Client → Amazon S3 (Frontend) → API Gateway → AWS Lambda → Amazon DynamoDB

Appendix B – Facial Recognition Attendance Architecture

The Facial Recognition Attendance System uses Amazon S3 for image storage and AWS Lambda for processing. Amazon Rekognition performs facial analysis, while DynamoDB is used to store employee and attendance-related information. API Gateway and EventBridge support application integration and event-driven communication.

Architecture Flow:

Image Source → Amazon S3 → AWS Lambda → Amazon Rekognition → Amazon DynamoDB

Supporting Integration:

API Gateway + EventBridge

Appendix C – AWS LMS / Certificate Generation Workflow

The supporting LMS cloud workflow involved AWS Lambda functions for processing, Python-based serverless logic for certificate generation, and Amazon S3 for storing generated certificates.

Architecture Flow:

LMS Application → AWS Lambda → Certificate Generation → Amazon S3

Supporting Services:

Amazon CloudWatch for monitoring and troubleshooting, with appropriate IAM permissions for secure access to AWS resources.

Appendix D – AWS Services Used During Internship

The following AWS services were used or explored during the internship:

AWS Service	Purpose / Application
Amazon EC2	Cloud computing and virtual-server concepts
Amazon S3	Frontend hosting, object storage, images and documents
AWS Lambda	Serverless application logic and processing
Amazon API Gateway	HTTP API and application integration
Amazon DynamoDB	NoSQL database and data persistence
Amazon Rekognition	Facial analysis
Amazon EventBridge	Event-driven integration
Amazon SNS	Notification workflows
Amazon SES	Email-related workflows
Amazon CloudWatch	Monitoring, logging and troubleshooting
AWS IAM	Roles, policies and access permissions

This appendix provides a consolidated overview of the major AWS architectures and services involved in the internship.